Object Oriented Programming

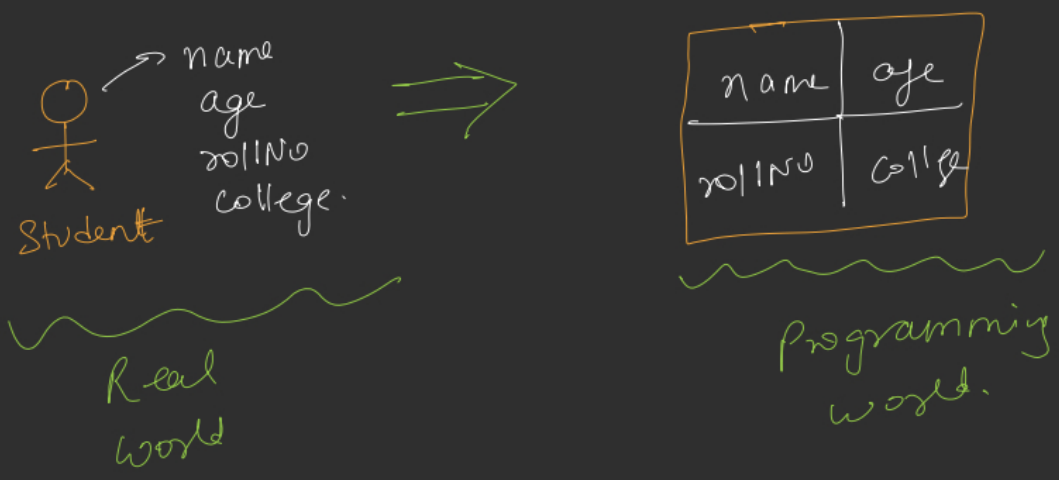
```
String name = "Aditya";
int age = 28;
int rollNo = 101;
String College = "IIT Guwahati";

String name2 = "Rohit";
int age2 = 28;
int rollNo2 = 102;
String College2 = "IIT Guwahati";
```

```
void print(String name, int age, int rN, int college){
    System.out.println( ——— );
}
```

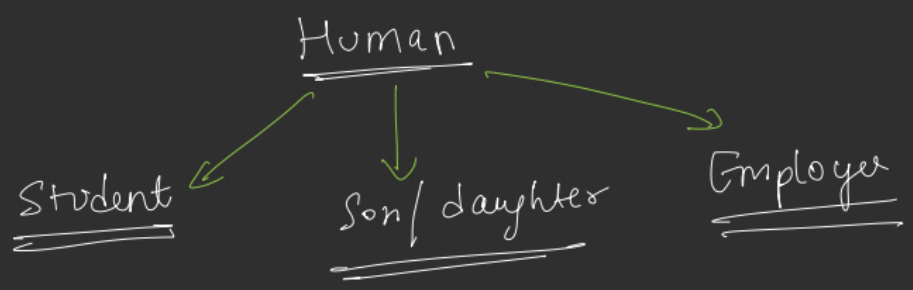
Problems

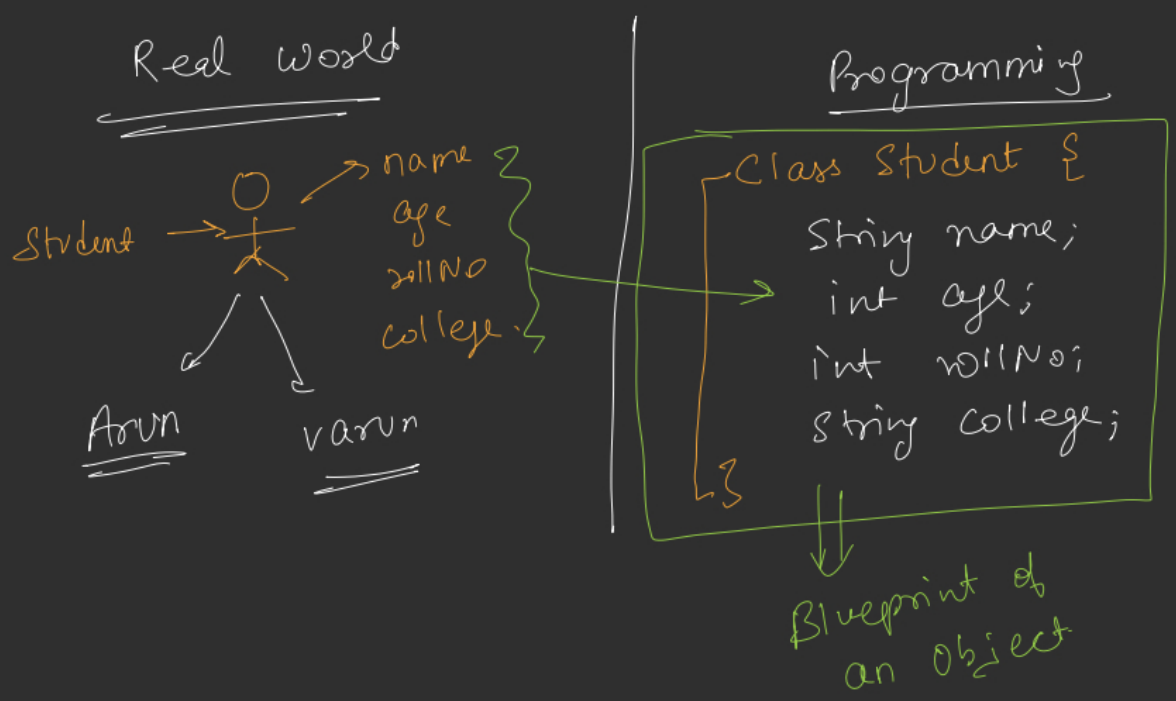
- Need a lot of independent variable to represent one student.
- If I pass the student to a function I need to pass all 4 variables.
- I need to create new set of variables for new student.
- No authority over data.

Object Oriented Programming



{ name, age, rollNo, college }





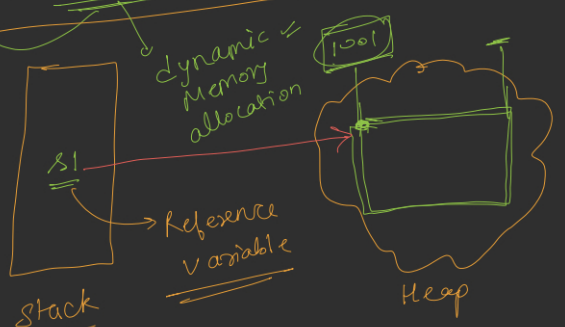
```
Class Student {
    String name;
    int age;
    int rollNo;
    String college;
}
```

DataType

- Primitives
  - Integer
  - Floating
  - Boolean
  - Character
- Non-primitives
  - ⇓ Objects
  - User-defined data Type

```
Student s1;
s1 = new Student();
```

dynamic Memory allocation



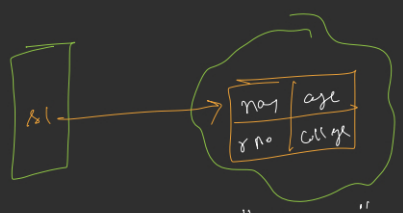
Run-time

→ Compile time (static)

→ Run time (dynamic)

```
Student s1 = new Student();
```

```
Class Student {
    String name;
    int age;
    int rollNo;
    String college;
}
```

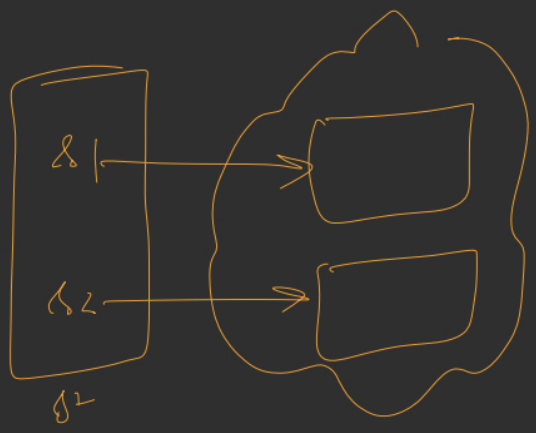



```
s1.name = "Aditya";
s1.age = 28;
s1.rollNo = 101;
s1.college = "IIT Guwahati";
```

```
Student s2 = new Student();
```

```
s2.name = "Rohit";
s2.age  = 28;
s2.r    = 102;
```

```
Class Student {
    String name;
    int age;
    int rollNo;
    String college;
}
```




```
Student s1 = new Student();
Student s2 = new Student();

s1.name = "Aditya";
s1.age  = 28;
s1.rollNo = 101;
s1.College = "IIT Guwahati";
```

System.out.println(s1.name);

```
s2.name = "Rohit";
s2.age  = 28;
s2.rollNo = 102;
s2.College = "IIT Guwahati";
```

Naming Convention (Good practice)

variables (primitive / reference variable)

↳ Camel Case

firstName, lastName

firstNameOfStudent

FirstName

student

Classes → Starting with Capital letter

Student, Human, Char

functions → Camel Case.

Student

→ Characteristics

→ Behaviours

```
Class Student {
  ≡
}
```

Animal

↳ Characteristics (Type, no of legs, ---)

↳ Behaviour (make sound, eat, run, ---)

```
Class Animal {
    string type;
    int noOfLegs;
    --
    void makeSound () {
        =
    }
    void eat () {
        =
    }
}
```

Student

Characteristics / data

↳ name
age
rollNo
college

Behaviour

↳ markAttendance

```
Class Student {
    string name;
    int age;
    int rollNo;
    string college;
    void markAttendance () {
        =
    }
}
```

Characteristics / data

Behaviour

```
void print (Student s){
    ——— (s.name, s.age --);
}
```

```
Student s1 = new Student();
s1.name;
s1.markAttendance();
```

```
Class Student {
    =
    =
    =
    void print () {
        ———
    }
}
```

Real Object	→	Digital Object
Animal	→	✓
Student	→	✓
BankAccount	→	✓
Location	→	✓

↳ long
↳ lat